

Sunil kumar Kommineni

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Professional Summary

Analyst & Engineering Data Science Master’s Candidate with proven success in optimizing decision-making, improving data quality and building predictive models. Skilled in Python, SQL, Tableau, Power BI and AWS, with experience delivering data-driven solutions for finance, education, healthcare and IoT. Adept at translating complex datasets into actionable insights that accelerate business outcomes.

Education

University of Houston, Cullen College of Engineering	May 2025
Master of Science Engineering Data Science GPA: 4.0 / 4.0	
Coursework: Machine Learning, Data Analytics, AI for Engineers, Database Management, Statistics	
National Institute of Technology, Puducherry	May 2023
Bachelor of Technology Electrical & Electronics Engineering GPA: 3.74 / 4.0	

Skills

Programming Languages: Python, SQL, R
Data Visualization Tools: Power BI, Tableau, Microsoft Excel, Matplotlib, Seaborn
Frameworks and Libraries: TensorFlow, PyTorch, Keras, Numpy, Pandas, Scikit-learn
Others: Amazon Web Services (AWS), Git, Jira, MongoDB, Data Wrangling, LLMs, AI, Prompt Engineering, NLP

Work Experience

Graduate Assistant - Team Lead	November 2023 – May 2025
University of Houston	Houston, TX
- Analyzed and optimized scheduling data for 5,000+ students per semester, leveraging data analysis techniques in Excel to ensure conflict-free exam schedules and enhance allocation efficiency. - Led a team of 20 Graduate Assistants to standardize data entry and improve accuracy for exam records by implementing automated Excel functions and reducing data entry errors by 40%.	
Student Intern	July 2021 – August 2021
Verzeo	Bengaluru, India
- Designed and developed an IoT-based soil monitoring system using Tinkercad and Arduino, integrating sensors to track moisture and temperature, enabling real-time monitoring with 95% data accuracy. - Implemented live data collection and visualization on ThingSpeak, handling updates every 15 seconds, improving environmental insight and system responsiveness.	

Projects

Advanced Facial Recognition and Bias Mitigation	September 2024 - November 2024
- Processed and balanced a subset of 8,000 images from the FairFace dataset to ensure inclusive representation across 7 racial and 8 age groups with the ETL process. - Performed exploratory data analysis (EDA), feature selection and improved model performance by 10–15% through regularization and hyperparameter tuning. - Developed a fairness-aware facial recognition system using EfficientNetB0 achieving 85% accuracy in gender prediction.	
Bank Loan Performance Analysis and Risk Profiling	February 2024 - May 2024
- Analyzed 10,000+ loan applications using KPIs like Debt-to-Income ratio, interest rates, and repayment history to identify high-risk borrower segments, improving loan approval accuracy by 15%. - Created an interactive Tableau dashboard to visualize MTD vs PMTD loan metrics, loan status distribution and enabling senior leadership to make data-driven lending decisions 20% faster. - Collaborated with domain experts to define rules for good vs. bad loans and implemented risk profiling logic, which led to a 30% decrease in potentially defaulting loan approvals.	
Drug Consumption Analysis & Prediction	September 2023 - November 2023
- Cleaned and transformed 1,800+ drug consumption records to address missing values, outliers, and inconsistencies, resulting in an analysis-ready dataset for behavioral insights. - Conducted exploratory data analysis using 15+ visualizations and statistical tests to uncover key patterns in substance use behavior and the development of predictive models with >80% accuracy. - Engineered 10+ features and applied normalization techniques to prepare data for machine learning, leading to a 25% improvement in model performance metrics.	